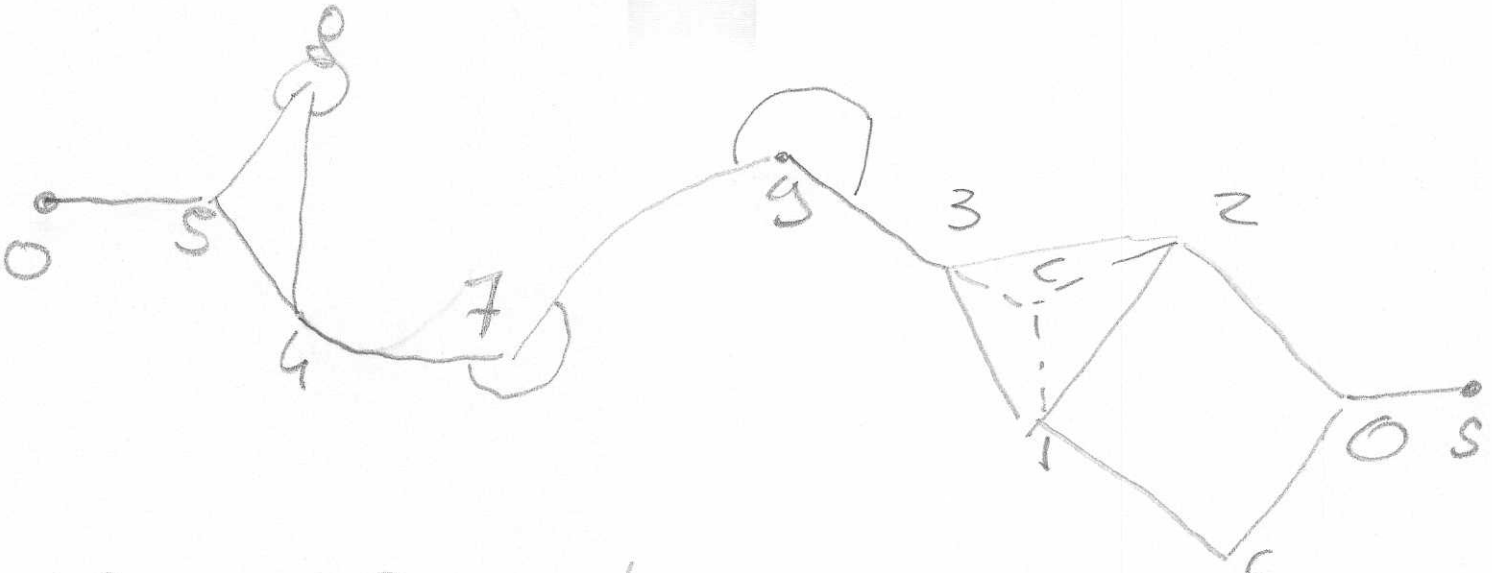
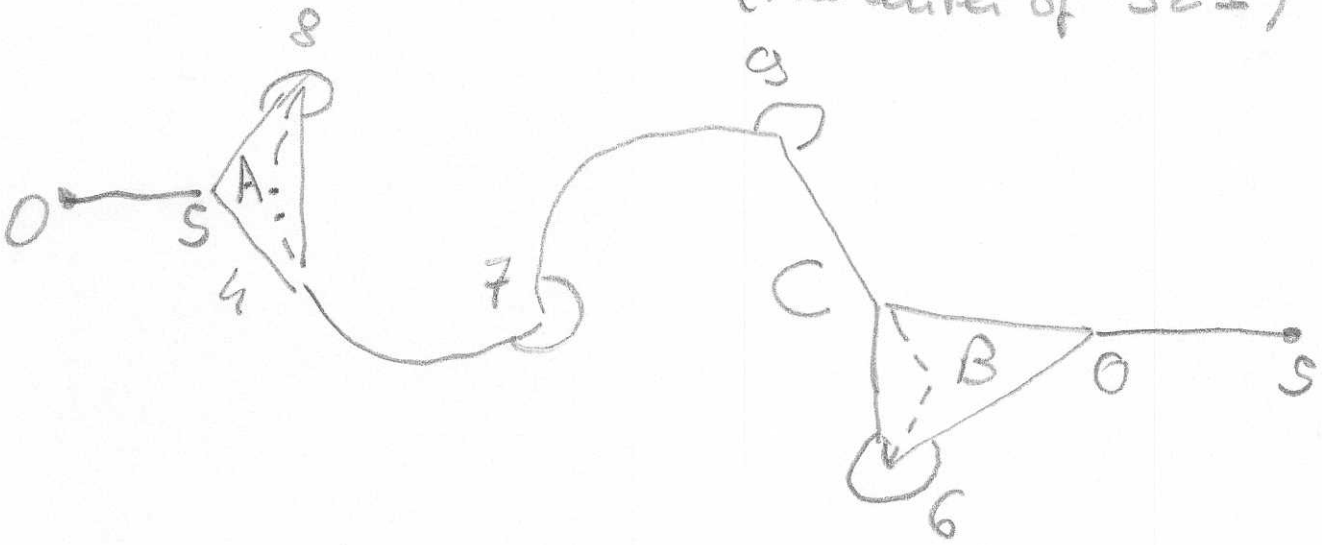
2834

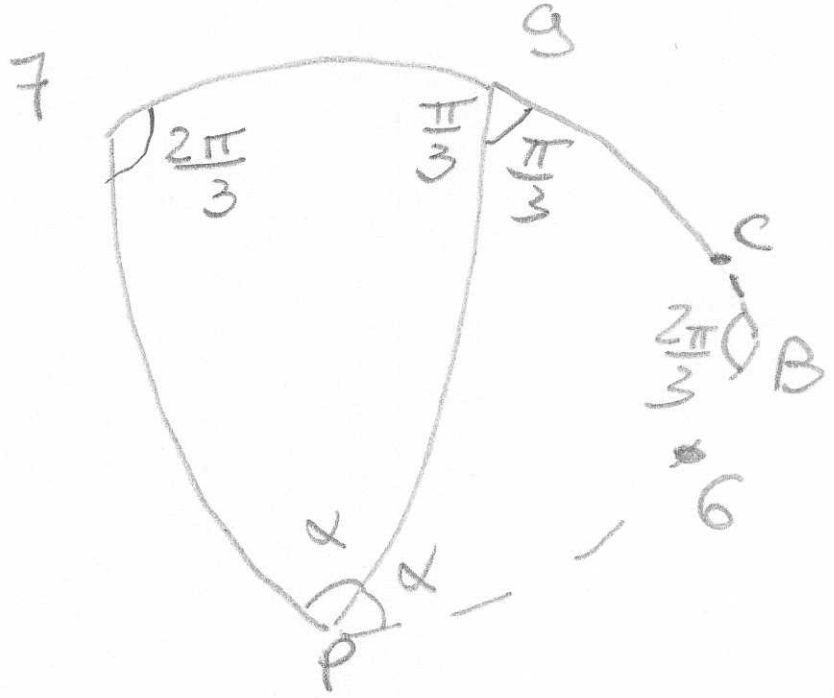


We apply the triangle elimination and replace the vertices 3, 2, 1 with C (the center of 321)



We continue the bisector of 6 and the edge 9B until they meet in the center of the triangle 6C0

We connect 7, 9 and B to one of the pins



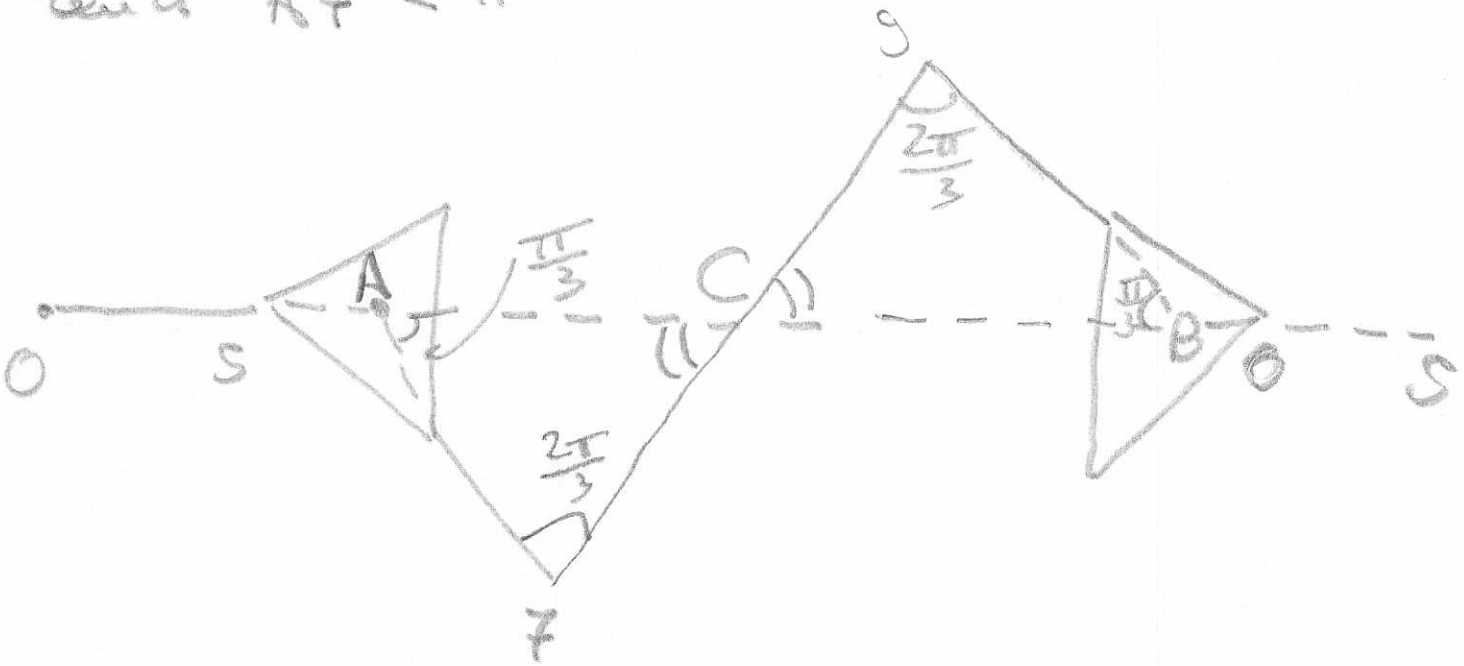
The triangles 79P and 9BP are congruent. In particular they must both be convex ($\alpha < \pi$) and so 9B and 79 are shorter than π.

The same argument applies symmetrically using the triangle 56P

Now O, S, A (the center of (845)) and B are all on the same equator, the bisector of O and S.

We let H^+ and H^- be the hemispheres bounded by γ

$9 \in H^+$ and $7 \in H^-$ because $\overline{B9} < \pi$ and $\overline{A7} < \pi$



Let C be the intersection of the equator γ with $\overline{79}$.

The triangles A7C and C9B are congruent (the angles are equal)

So $\overline{FC} = \overline{CG} = \frac{1}{2}\overline{FG} = \frac{1}{2}\overline{GB}$

We can now determine the triangle CGB

If we set $\overline{CG} = a$, then $\overline{GB} = 2a$

and $\overline{CB} = \pi - a$ (because $\hat{G} + \hat{B} = \pi$)

So

$$\cos a = \cos 2a \cos(\pi - a) + \sin 2a \sin(\pi - a) \cdot \cos\frac{\pi}{3}$$

$$\cos a = (2\cos^2 a - 1)(-\cos a) + \sin^2 a \cos a$$

$$\cos a = 1 - 2\cos^2 a + 1 - \cos^2 a$$

$$\cos^2 a = \frac{1}{3}$$

$$\cos a = \sqrt{\frac{1}{3}}$$

(because $a < \frac{\pi}{2}$)

But $BO = a\cos\frac{1}{3}$

So

$$2\pi > \overline{SA} + \overline{BO} + \overline{CB} + \overline{CA}$$

$$= 2a\cos\frac{1}{3} + 2\pi - 2a\cos\frac{1}{\sqrt{3}}$$

which would imply

$$a\cos\frac{1}{\sqrt{3}} > a\cos\frac{1}{3}$$

$$\text{i.e. } \frac{1}{\sqrt{3}} < \frac{1}{3}$$